

ALB + ASG

Causal chain

- EC2 health detects the instance is gone.
- Sequence:
- Instance terminated
 - EC2 reports instance unhealthy / gone
 - ASG notices desired capacity violated
 - ASG launches replacement
- Target Groups are NOT involved when you Kill an instance

Health check path

- App stops responding on /health (or /)
- Target Group marks instance Unhealthy
- ALB stops routing traffic
- ASG (with ELB health enabled) sees Unhealthy
- ASG terminates instance
- ASG launches new instance
- New instance also fails health check
- Loop repeats forever

This is called health-check-induced down.

In production, this is a real outage mode and why health checks must be stable and simple.

Traffic routing behavior

- ALB distributes traffic across healthy targets
- When multiple healthy targets exist → round-robin (simplified)
- When only one exists → all traffic to it
- When none exist → ALB returns 503

What load balancers actually solve

1. Traffic distribution
2. Failure isolation
3. Zero-downtime replacement
4. Health-based routing

Traffic splitting is the least important one in production.

Why ALB health $\neq$ instance health

ALB health checks are HTTP based and they check if an app is running properly. EC2 instance health checks simply determine if an instance is running or not.

Why "just use a bigger server" fails in the cloud

Because of failure isolation concerns. One DDoS attack could overwhelm your application if the back bone is just one server. You want your architecture on prod to be highly available and redundant to mitigate these concerns.

Why ALB vs Network LB vs Gateway LB?

An ALB is best when you're running a typical web application using HTTP or HTTPS. It works at the "request" level, meaning it understands individual web requests and can make smarter routing decisions. This is useful when you need features like sending different URLs to different services, and it's especially helpful for modern setups like microservices and containers.

A NLB is the best choice when you need extremely high performance and very low delay. It works at the "connection" level, meaning it focuses on moving traffic as fast as possible instead of analyzing each request. It can handle millions of requests per second, supports UDP traffic, can provide static IP addresses, can perform TLS offloading, which means it can handle encryption and decryption at scale for secure traffic.

A GWLB is mainly used for security and network control. It's designed for deploying and managing fleets of third-party virtual appliances (like firewalls or inspection tools) that support GENEVE. These appliances help organizations improve security, compliance, and policy enforcement by making sure traffic is inspected or processed before reaching applications.